

Protection

When tanks were conceived, they were imagined to have one sole function: to precede attacking infantry across no-man's-land and give them protection from enemy machine gun fire by supressing it with their own guns and machine guns. They themselves had to be protected, which meant equipping them with 0.47 in (12 mm) of rolled steel armor on their exposed front faces, although that soon increased to 0.55 in (14 mm) to withstand the armor-piercing 7.92 mm K bullet.

However, it was not possible to make armor thick enough to protect against the German 7.7 cm field gun, which was soon in an antitank role. By the 1930s, effective antitank guns had also appeared—and had of course found their way into tanks. Thus a vicious circle was established, with ever more powerful antitank guns being created and put into tanks, and designers piling heavier and heavier armor onto their vehicles in the hope, often forlorn, of staying ahead of the opposition.

ARMOR

The earliest type of armor consisted of plates of rolled steel, which were made by passing cast billets between rollers until the metal was the desired thickness. This repeated compression had the effect of aligning the molecules in the steel, which toughened the material. The next stage was face hardening, which saw the plates reheated on a bed of granular carbon, a process known as “carburizing” (the two types were often employed together, to produce what was known as “cemented” armor—a process developed in Germany by Krupp). From then on it was necessary to introduce alloys such as chromium, molybdenum, nickel, and later tungsten to produce a tougher product. Some antitank rounds burn through armor rather than penetrating by kinetic energy and to combat these, layers of ceramic blocks were introduced, giving modern vehicles their distinctive angular appearance. Such armor is often known as “Chobham,” after the Surrey town where it was developed, and is invulnerable to AT rounds.



Light armor
Small, light tanks such as the British Mark VIB sacrificed armor weight for the sake of speed, maneuverability, and transportability.



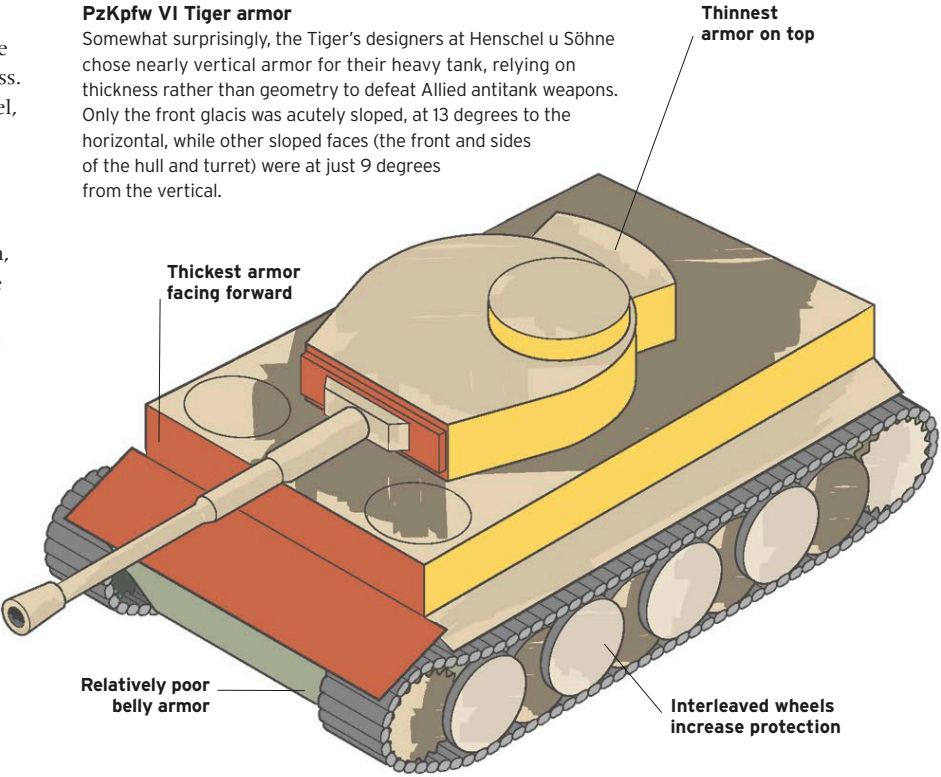
Heavy armor
Large, heavy vehicles such as the German Jagtigger sacrificed speed and maneuverability for the sake of protection.



Composite armor
Modern vehicles such as the Israeli Merkava Mark 4 are both fast and maneuverable, being protected by composite armor, which is generally lighter than all-metal alternatives.

PzKpfw VI Tiger armor

Somewhat surprisingly, the Tiger's designers at Henschel u Söhne chose nearly vertical armor for their heavy tank, relying on thickness rather than geometry to defeat Allied antitank weapons. Only the front glacis was acutely sloped, at 13 degrees to the horizontal, while other sloped faces (the front and sides of the hull and turret) were at just 9 degrees from the vertical.



25 mm 60 mm 80-100 mm 100-120 mm

SLOPED AND UNSLOPED ARMOR COMPARED

Setting armor at an angle to the vertical offers two advantages. Firstly, the angle increases the thickness of the armor to be penetrated. Secondly, it makes it likelier that an antitank projectile, especially one with an curved profile, will be deflected away from the tank and so expend itself uselessly.

